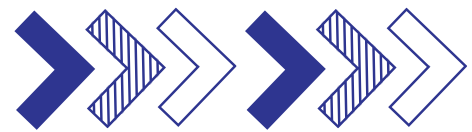




OPEN IIT DATA ANALYTICS

2024

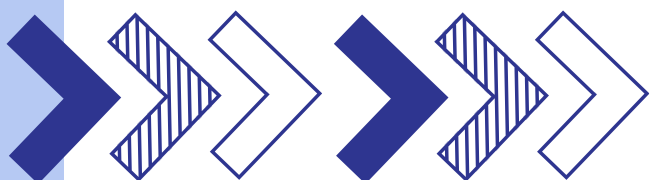


TEAM D-33



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PROBLEM STATEMENT

As air travel becomes increasingly popular, the frequency of flight delays has also risen, leading to greater congestion at airports and financial challenges for airlines. Flight delays serve as indicators of inefficiencies within the aviation system, resulting in substantial costs for both airlines and passengers. Effectively predicting flight delays is crucial for enhancing airline operations, improving passenger satisfaction, and positively impacting the broader economy.

This initiative focuses on two main components:

- **Predictive Modeling:** Analyzing historical flight data—like schedules, weather, and crew availability—to forecast potential delays, helping airlines manage resources effectively.
- **Optimized Flight Rescheduling:** Developing strategies to reschedule flights based on the given set of delays, considering factors such as aircraft availability and crew schedules to minimize disruptions.

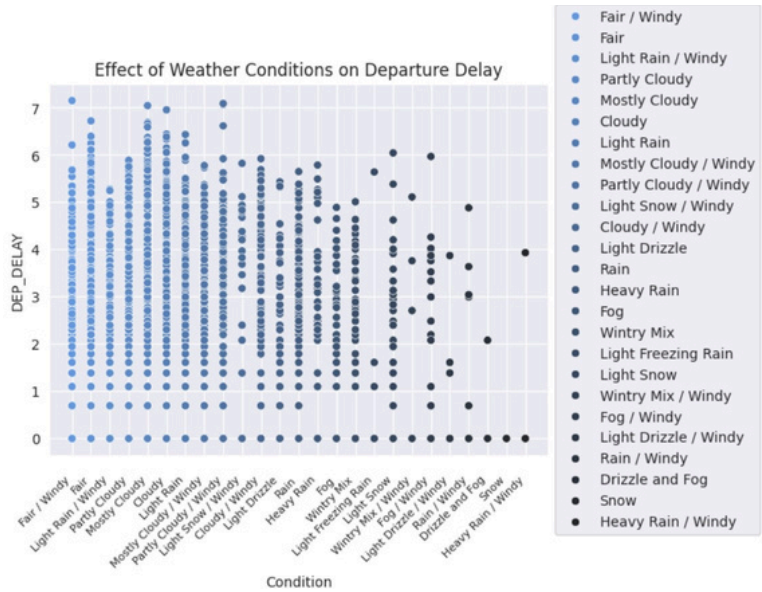
Implementing this predictive model and rescheduling system will improve operational efficiency and customer satisfaction by reducing delays and ensuring timely services.

BACKGROUND

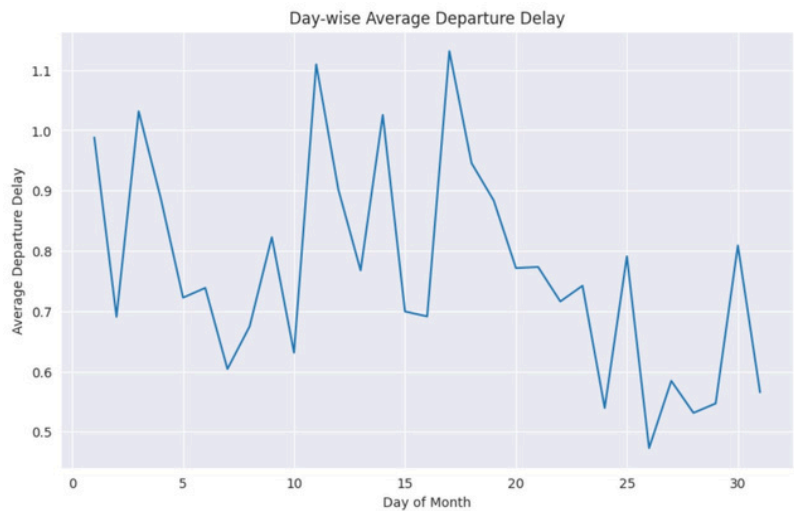
In this analysis, we employed a predictive model to assess delay factors within the dataset, focusing on variables like staffing levels, passenger flow, and air traffic volume. Using these features, the model was trained to recognize patterns and predict potential delays, helping to identify how variations in each factor contribute to scheduling disruptions. This predictive approach allows for proactive rescheduling and efficient delay management, providing insights that support better decision-making for operational resilience and smoother scheduling adjustments. By analyzing these dynamics, we aim to reduce delays and enhance overall schedule reliability.

EXPLORATORY DATA ANALYSIS

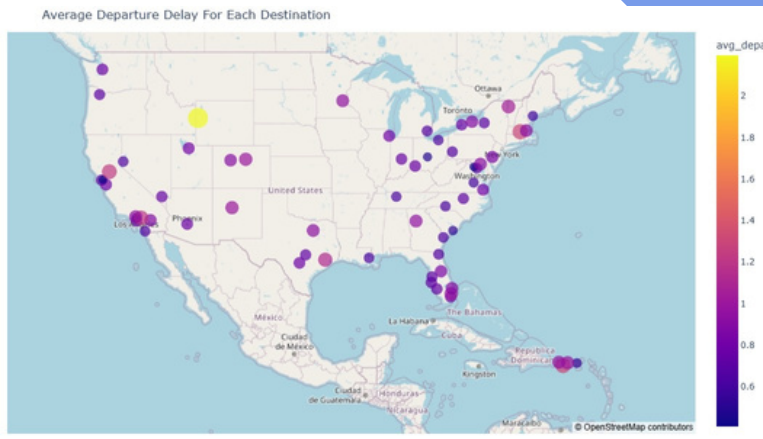
This dot plot shows the distribution of departure delays across various weather conditions, highlighting patterns, variations, and outliers within each category. Severe weather, like heavy rain, fog, and wind, is associated with higher delays, while fair or partly cloudy conditions have minimal impact.



The line graph depicts daily average departure delays for flights over a month, highlighting significant variability in punctuality. Peaks and troughs indicate that factors such as weather, operational issues, and demand fluctuations impact flight schedules differently each day. This analysis emphasizes the complexities of maintaining timely departures in air travel.

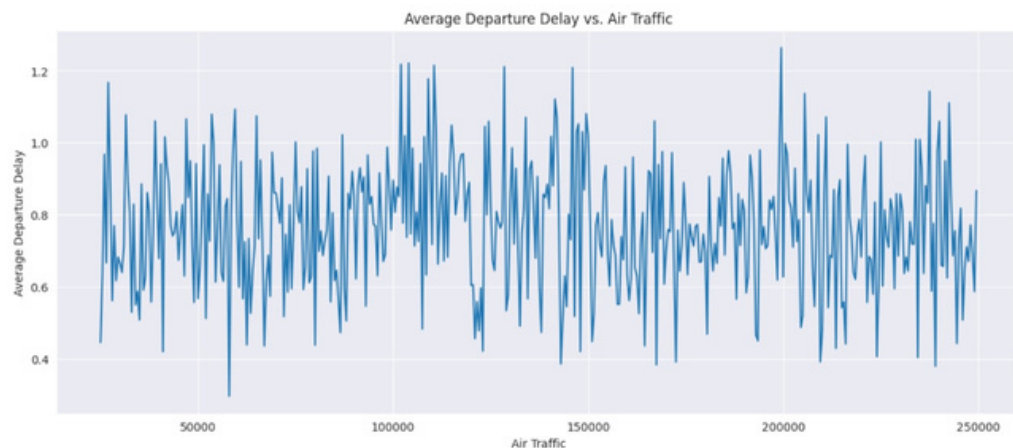
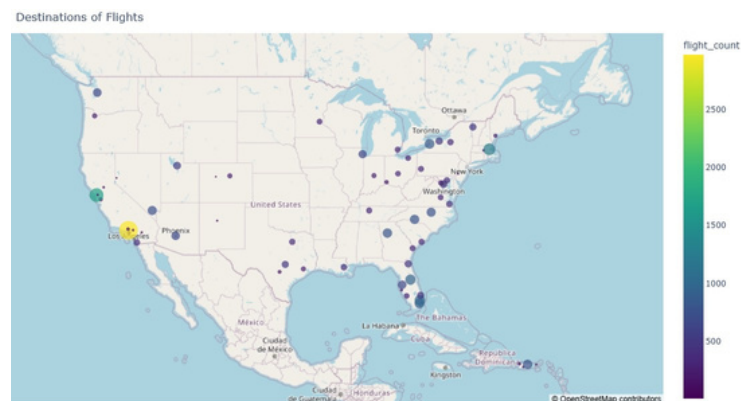


The point-based choropleth map illustrates average departure delays for each destination, with circle size and color intensity (shifting from blue to yellow) indicating longer delays. This visualization helps identify cities with consistently high delays, suggesting potential systemic issues in airport management or air traffic congestion.

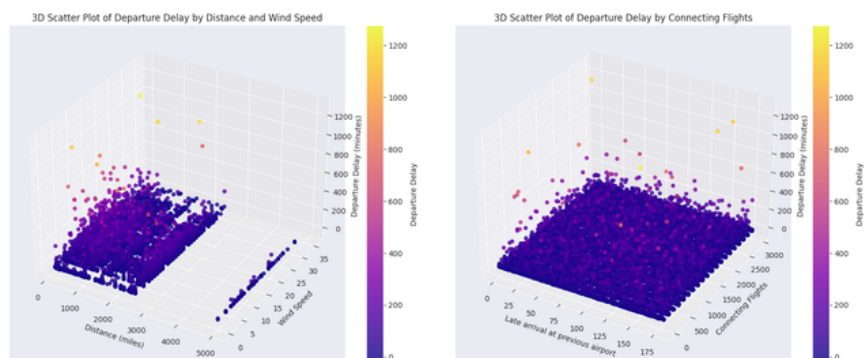


EXPLORATORY DATA ANALYSIS

This point-based choropleth map illustrates the geographic distribution of flight destinations from a specific airport or group of airports. Each point represents a destination city, with circle size and color intensity indicating flight volume, where larger and more vibrant circles signify higher traffic. This visualization provides insights into the most trafficked routes and destinations.



The line graph depicts the correlation between air traffic volume and average departure delay, with each point representing specific traffic levels and their corresponding delays. The lack of a clear linear relationship indicates that factors such as weather conditions, runway availability, and air traffic control management may significantly influence departure punctuality, potentially more so than traffic volume alone.



The two 3D scatter plots analyze factors affecting flight delays. The first plot shows that longer flight distances and higher wind speeds correlate with increased departure delays, as indicated by the concentration of yellow points in the upper right region. The second plot demonstrates that a greater number of connecting flights and longer delays from previous flights are associated with longer departure delays for the current flight, evidenced by the clustering of yellow points in areas with higher connecting traffic and previous delays.

DATASET JUSTIFICATION

This dataset provides a comprehensive view of the factors affecting flight delays, allowing us to build an effective predictive model. Below is a breakdown of how key factors contribute to flight delays and rescheduling challenges.

FACTOR CATEGORY	KEY VARIABLE (COLUMNS)	INSIGHTS
Temporal Factors	MONTH, DAY_OF_MONTH, DAY_OF_WEEK	Winter months face weather disruptions; end-of-month increases traffic; weekdays have higher congestion, weekends may lack staff.
Flight & Carrier-Specific Factors	OP_UNIQUE_CARRIER, TAIL_NUM, CONNECTING_FLIGHTS	Different carriers have varying efficiencies; connecting flights can cause cascading delays.
Airport & Ground Operations	AIR_TRAFFIC, GROUND_STAFF_AVAILABLE, FUEL_AVAILABILITY, LATE_ARRIVAL_PREV_AIRPORT	Congestion and staff shortages delay handling, while arriving delays disrupt turnaround times.
Passenger Behavior	FOOTFALL_PASSENGERS, PERC_PASSENGERS_LATE	High passenger numbers slow security checks; late arrivals disrupt schedules.
Weather & Environmental Conditions	TEMPERATURE, DEW_POINT, HUMIDITY, WIND, WIND_SPEED, WIND_GUST, CONDITION	Adverse conditions delay take-offs/landings; extreme weather complicates schedules.
Flight Stage-Specific Factors	DEP_DELAY, TAXI_OUT, CRS_ELAPSED_TIME, DEP_TIME_M, CRS_DEP_M	Identifying delays during departure and taxiing can help optimize schedules.



DATA PREPROCESSING

Missing Values

- **Changing NaN to Median:** Values that were NaN were changed to the median of that column. This method was chosen to keep the statistical validity, especially in columns where exceptions could change the mean. Keeps things consistent and lessens the effect of missing numbers on statistical measures as much as possible.
- **Replacing NaN with Zero:** Values that were NaN were replaced with zero in certain fields. This method was picked for variables where NaN means that the trait being measured is not present. By giving a value of zero, the lack of data is correctly shown, which makes analysis and understanding easier.
- **Reason for Choice of Method:** The features of the dataset led to the choice to use these methods. We selected Median to replace the NaN values because the data would not get affected by outliers.

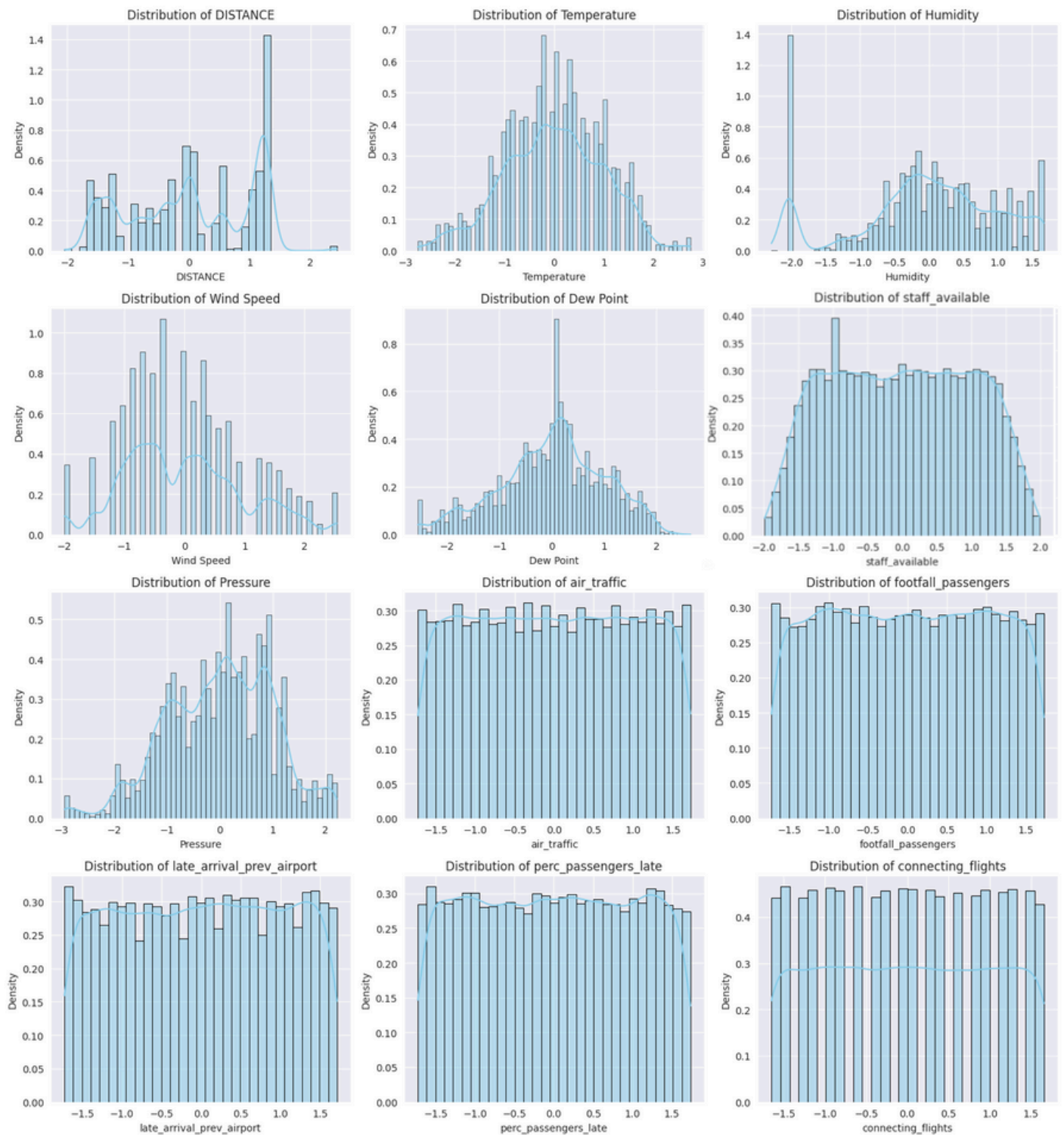
Label Encoding

- **Application in Our Dataset:** In our analysis, label encoding was applied to categorical columns such as OP_UNIQUE_CARRIER, Wind, and Condition. This conversion enabled these features to be included in the model training process, ensuring the algorithm could interpret the data correctly.
- **Benefits-**
 - 1.Data Readiness:** Transforms categorical data into a format suitable for machine learning, allowing for the utilization of all relevant features.
 - 2.Enhanced Model Performance:** By encoding these variables, we potentially improve the model's predictive accuracy.
 - 3.Ease of Implementation:** The method is straightforward to implement using libraries like scikit-learn.

Normalization Methodology

Normalization Method	Skewness	Purpose
Z-Score Normalization	Close to 0 (Normal)	Transforms data to have a mean of 0 and SD of 1.
Yeo-Johnson Transformation	Moderately/Highly Positive or Negative	Normalizes data with both positive and negative values.
Log Transformation	Highly Positive	Compresses large values to reduce skewness.

VISUALIZATION OF SCALED DATA



OUTLIER DETECTION

Outlier Detection and Handling Using the IQR Method

The IQR method is a statistical technique used to identify outliers in a dataset. It involves dividing the data into quartiles: the first quartile (Q1) represents the 25th percentile, and the third quartile (Q3) represents the 75th percentile.

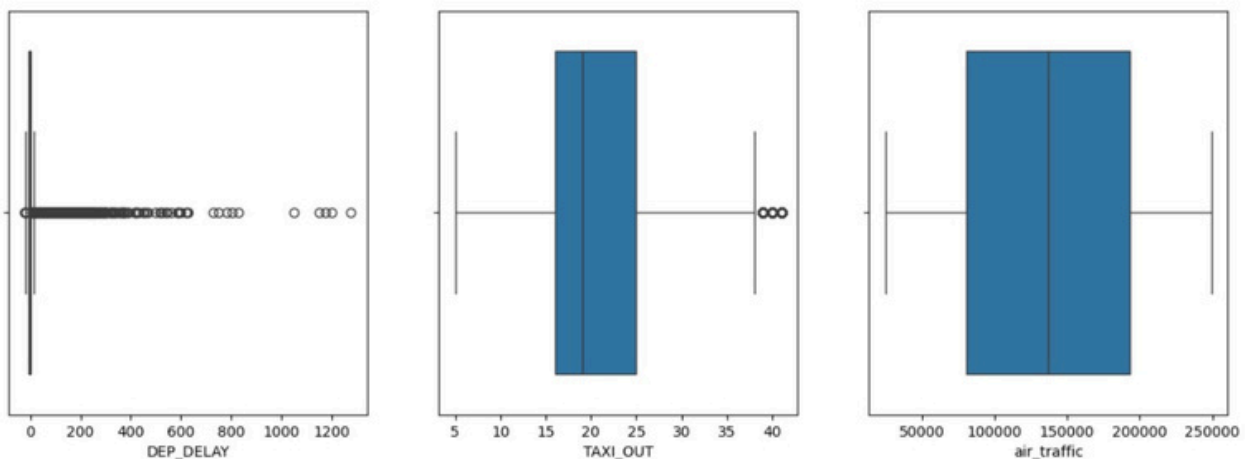
The Interquartile Range (IQR) is the difference between Q3 and Q1, measuring the spread of the middle 50% of the data. Outliers are defined as data points that fall below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. These values may represent errors, rare occurrences, or anomalies that deviate significantly from the normal distribution of the data.

Handling Outliers Through Capping:

Outlier handling by capping is a technique that modifies extreme values to reduce their influence on the dataset without fully removing them. This approach replaces outliers that lie beyond a certain threshold with the closest value within the accepted range, effectively “capping” these extreme data points.

Lower Bound Capping: This involves setting a minimum acceptable value. Any data points that fall below this threshold are replaced (or “capped”) by the lower bound value. For instance, if the lower bound is defined as $Q1 - 1.5 \times IQR$, any value below this limit is capped at that threshold, effectively preventing extremely low outliers from affecting the dataset.

Upper Bound Capping: Similarly, this sets a maximum acceptable value for the data. Data points above this threshold are replaced with the upper bound value, usually defined as $Q3 + 1.5 \times IQR$. This capping prevents extremely high values from skewing the data distribution.



PREDICTIVE MODEL

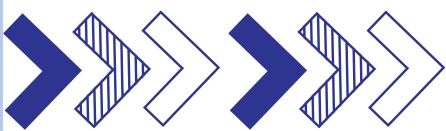
The predictive model is designed to optimize decision-making in airline operations by forecasting flight delays and enabling effective rescheduling. To handle the intricate nature of airline data, a set of regression models—XGBoost, LightGBM, and CatBoost—was employed, with each model fine-tuned through hyperparameter optimization to enhance its individual predictive accuracy. These models were then integrated using a Stacking Regressor, where a linear regression model serves as the meta-learner, aggregating outputs from each base model. This ensemble strategy enhances overall prediction accuracy by combining the strengths of each algorithm, thereby improving robustness and generalization across diverse operational conditions.

Individual Model Descriptions

- **XGBoost:** Known for its robust handling of structured data, XGBoost excels in high-dimensional spaces and performs well in noisy environments. Hyperparameters such as the number of estimators, learning rate, and maximum depth were fine-tuned, yielding an RMSE of approximately 11.18 on the test data.
- **LightGBM:** LightGBM was selected for its efficiency in handling large datasets and ability to maintain high accuracy. Hyperparameters like learning rate, max depth, number of leaves, and minimum child samples were optimized. LightGBM achieved a test RMSE of approximately 14.95.
- **CatBoost:** CatBoost, effective for categorical data, was tuned for optimal depth and learning rate. Although its individual performance showed an RMSE of around 15.46, its contribution was vital in ensemble learning.

Stacked Ensemble

To harness the collective strengths of these models, a Stacking Regressor was implemented. The stacking model combines predictions from each base model using a linear regression model as the meta-learner, which improves generalization and predictive accuracy. The stacked model significantly enhanced the performance metrics, achieving a mean absolute error **(MAE) of 3.04**, **root mean squared error (RMSE) of 12.29**, and an **R-squared value of 0.896**. **This high R-squared indicates that the model explains approximately 89.6%** of the variability in flight delay outcomes, highlighting its robustness for deployment across diverse airline operational scenarios.



EXPLANATION

The Stacking Regressor approach is used here to enhance predictive accuracy and robustness by combining multiple machine learning models—XGBoost, LightGBM, and CatBoost—each with distinct strengths suited for different aspects of complex airline data.

- **XGBoost** is well-suited for handling large datasets with high dimensionality, leveraging its ability to capture complex interactions through gradient-boosted trees.
- **LightGBM** offers fast training and high accuracy by employing leaf-wise growth, making it efficient for numerical features.
- **CatBoost** effectively handles categorical data, preventing overfitting on noisy datasets and maintaining stability even in heterogeneous data distributions.

By using Linear Regression as the meta-learner, we create a weighted combination of predictions from each base model. Linear Regression, being a simple model, acts as a “blending” layer, effectively combining the diverse strengths of the individual models without introducing extra complexity. This meta-learner enhances generalization and predictive accuracy, helping to reduce prediction variance by balancing out biases from each model.

Why Stacking Regressor?

- **Generalization to Diverse Scenarios:** The ensemble approach helps capture a broader range of data patterns, making it more adaptable to diverse operational scenarios (e.g., different airports, varying flight routes, and aircraft types). This is critical for the airline industry, where operations are highly variable.
- **Reduced Overfitting:** The use of multiple models reduces the likelihood of overfitting, as each base model brings a slightly different perspective, helping to smooth out predictions in the final model. Linear regression as the meta-learner balances these inputs without heavily weighting any one model, thus improving stability and reducing variance.
- **Improved Predictive Accuracy:** By leveraging the strengths of multiple models—XGBoost for handling high-dimensional data, LightGBM for efficiency with numerical features, and CatBoost for robustness with categorical data—the Stacking Regressor achieves a higher predictive accuracy than any individual model. This leads to more precise predictions on key metrics, such as flight delays, which can directly improve decision-making and operational efficiency in airline management.

EVALUATION METRICS	
MAE (Mean Absolute Error)	3.04
RMSE (Root Mean Squared Error)	12.29
R- Squared	0.896

HYPERPARAMETER TUNING

with RandomizedSearchCV

Hyperparameter tuning has been done to find the best combination of hyperparameters to optimise model performance for each algorithm.

Implementation:

Each model (**XGBoost**, **LightGBM**, and **CatBoost**) is defined with its corresponding parameter grid, which specifies potential values for hyperparameters such as:

- **n_estimators**: Number of trees in the ensemble (controls model complexity).
- **learning_rate**: Learning rate (controls how much the model adapts with each iteration).
- **max_depth, num_leaves, min_child_samples (for LightGBM), subsample (for XGBoost), and depth (for CatBoost)**: Parameters affecting the tree structure, which can reduce overfitting.

RandomizedSearchCV has been applied to each model, with:

- **n_iter=10**: Limits the search to 10 random combinations of parameters to save time compared to an exhaustive grid search.
- **scoring='neg_mean_squared_error'**: Uses negative mean squared error as the scoring metric to minimize prediction error.
- **cv=3**: 3-fold cross-validation to ensure models are tested on different parts of the dataset.

Each parameter has been selected to tune the complexity and adaptability of the model, balancing accuracy and overfitting. These hyperparameters control the depth, learning rate, and amount of data used, making the models robust for predicting flight delays in varied conditions.

Model	Hyperparameter	Value
XGBoost	• subsample • n_estimators • max_depth • learning_rate	• 1 • 200 • 10 • 0.1
LightGBM	• num_leaves • n_estimators • min_child_samples • max_depth • learning_rate	• 50 • 50 • 10 • 10 • 0.2
CatBoost	• n_estimators • learning_rate • depth	• 150 • 0.1 • 10

FLIGHT RESCHEDULING

OBJECTIVE

The primary objective of this part is to reschedule flights to minimize departure delays by adjusting flight schedules based on weather, staffing availability, and distance thresholds. Specifically, it focuses on flights with high delays and attempts to reschedule them when favorable conditions are met, thus reducing overall delay across a set of flights.

DELAY MULTIPLIERS

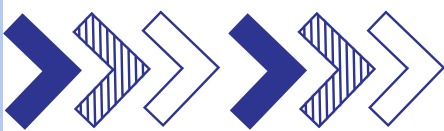
The delay multipliers function is designed to compute delay multipliers based on historical flight delay data. These multipliers are essential for adjusting departure times during the flight rescheduling process.

Parameters:

- **Data:** A pandas DataFrame containing flight data, including delay reasons, staff availability, distance, and departure delay.

Methodology:

- **Grouping by Factors:** The function groups the dataset by several key variables:
 - Condition: The reason for the flight's delay.
 - air_staff_available: The number of air staff available at the time of departure.
 - ground_staff_available: The number of ground staff present.
 - DISTANCE: The distance of the flight.
- **Calculation of Mean Delays:** For each group, the mean of the DEP_DELAY (departure delay) is calculated. This mean serves as the basis for deriving the delay multipliers.
- **Delay Multiplier Calculation:** The delay multiplier for each group is computed using the formula:
$$\text{Multiplier} = 1 - \left(\frac{\text{Mean Delay of Group}}{\text{Overall Mean Delay}} \right)$$
- Here, the overall mean delay is computed across all records in the dataset, thus providing a normalization factor.
- **Handling Missing Data:** In scenarios where certain combinations yield no data (i.e., when no flights match the group), the multipliers are assigned a default value of 0.5. This ensures continuity in the rescheduling algorithm, allowing for all conditions to be accounted for.
- **Return Value:** The function returns a dictionary mapping each unique combination of factors to its corresponding multiplier, facilitating quick lookups during the rescheduling process.



CONSTRAINTS

1. Weather Conditions:

- The code checks if the weather condition at the departure airport is favorable for the rescheduled flight.
- This ensures that the flight can take off and land safely without being affected by adverse weather.

2. Distance:

- A distance threshold is defined to limit the extent of rescheduling.
- This prevents rescheduling flights to airports that are too far away, which could lead to longer travel times and inconvenience for passengers.

3. Staff Availability:

- The code checks if there are sufficient air and ground staff available at the departure and arrival airports to handle the rescheduled flight.
- This ensures smooth operations and avoids potential delays due to staffing shortages.

4. Aircraft Availability and Conflict Avoidance:

- The code verifies that the aircraft assigned to the flight is available at the new departure time.
- It also checks for potential conflicts with other flights using the same aircraft.
- If a conflict is detected, the departure time is adjusted to avoid overlapping schedules.

THRESHOLDS AND QUANTILES

In the code, thresholds are used to:

- **Filter flights for rescheduling:** Only flights that meet specific criteria (e.g., delay severity, weather conditions, staff availability, etc) are considered for rescheduling.
- **Optimize rescheduling:** Different threshold combinations for distance, air staff, and ground staff are tested to find the optimal configuration that maximizes delay reduction.

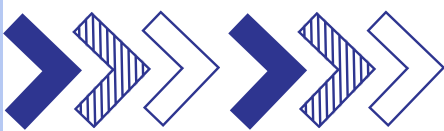
Quantiles are used to determine the threshold values for distance, air staff, and ground staff:

- **Calculate Quantiles:** The code calculates the 25th, 50th, and 75th percentiles for each of these variables.
- **Set Thresholds:** These quantiles are used as potential threshold values to filter flights and optimize rescheduling.

By using quantiles to set thresholds, the code ensures that the rescheduling process is adaptive to the specific characteristics of the dataset. This approach helps to avoid overly restrictive or permissive thresholds, leading to more effective rescheduling decisions.

Why quantiles for thresholds?

We've chosen a flexible approach, allowing each company to manually set thresholds to suit their unique operational needs, as these can vary widely across organizations. This method empowers companies to fine-tune rescheduling parameters for optimal alignment with their specific requirements, ensuring relevance and adaptability without rigid, one-size-fits-all thresholds.



METRICS

Optimal thresholds	
Distance	2248
Air Staff	6
Ground Staff	22
Maximum Delay Reduction Achieved	8800.5

APPENDIX

Evaluation Metrics

```
Tuned XGB Model RMSE: 11.176932967886843
Tuned LGBM Model RMSE: 14.949364872066537
Tuned CatBoost Model RMSE: 15.458336150586831
Stacked Model MAE: 3.0399036871243093
Stacked Model MSE: 150.92638220113878
Stacked Model RMSE: 12.285209896503144
Stacked Model R-square: 0.8964742904331356
```

Datasets

Predictive Analysis:

- 1.Train dataset: [Link](#)
- 2.Test dataset: [Link](#)

Flight Rescheduling:

- 1.Dataset: [Link](#)

